**PRESIDENT'SOFFICE**

**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**

**FORM FOUR MOCK EXAMINATION**

**CHEMISTRY 1**

**TIME:3: HRS** **MAY 2023**

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**INSTRUCTIONS**

- This paper consists of sections **A, B** and **C** with a total of fourteen (**11**) questions
- Answer all questions in section **A** and **B** and one (**02**) question from section **C**
- The following constants may be used:
- Atomic masses: **H=1, 0=16, Na=23, C=12, Cl=35.5, N=14, Zn=65, S=32, Ca=40, Fe=56, Cu=64, M=27**
- **GMV at S.t.p= Avogadro's number =**

**1 Faraday = 96500 Coulombs**

## SECTION A(16)MARKS

1. For each of the items (i)-(x) ,choose the correct answer from the alternatives and write its letter beside the item number in the answer sheet provided

(i) Which of the following sets of laboratory apparatus are used in measuring volume

a) Crucible, U-tube and volumetric flask
b) Test tube , Beaker and gas jar
c) Thistle funnel ,separating funnel and beaker
d) Burette, beaker and measuring cylinder
e) Conical flask ,test tube and measuring cylinder

(ii) The empirical formula of a certain compound is $CH_3$.Its molar mass is 30g what will be its molecular formular

a) CH4
b) C 2H4
c) C 2H6
d) C 2H8
e) C4H12

(iii) In order to produce the greatest amount of hydrogen in a short time one gram of magnesium ribbon should react with .

a) 10cm3 of 0.5 M sulphuric acid
b) 40cm3of 0.5M acetic acid solution
c) 40cm3of 0.5M sulphuric acid solution
d) 20cm3of 1M sulphuric acid solution
e) 20CM3 of 1M acetic acid solution

(iv) Fractional distillation prosses of a mixture of water and ethanol is possible because

a) Water and ethanol have the same boiling point
b) Water has lower boiling point than ethanol
c) Ethanol has lower boiling point than water
d) Water and ethanol form partially miscible liquid solution
e) Water and ethanol are immiscible liquids

(v) Which of the following substance represent a group of acid oxides?

a) Carbon dioxide ,carbon monoxide and sulphur dioxide
b) Sulphur trioxide ,nitrogen dioxide and nitrogen monoxide
c) Carbon dioxide ,sulphur dioxide and dinitrogen oxide
d)Sulphur trioxide ,carbondioxide and nitrogen dioxide
e) Carbon monoxide ,nitrogen oxide and sulphur dioxide

(vi) What will be the molarity of a solution which contains 26 .5g of anhydrous sodium carbonate in 5dm3 of solution ?
a) 0.05M
b) 0.25M
c) 5.30M
d) 0.025M
e) 0.50M

(vii) The Brownian movement is taken to be the evidence of the
a) Theory of association of water molecules
b) Theory of ionization of electrolytes
c) Theory of colloidal suspensions
d) Kinetic theory of behaviour of substances
e) Brownian theory

(viii) One of the isotopes of an element x has an atomic number z and a mass number A .what is the number of neutrons contained in the nucleus of
a) Zb) A
c) A+Zd) A-Ze) Z-A

(ix) C2H4Cl can be represented in different structure which are called
a) Homologous series
b) Isomers
c) Identical formulae
d) Identical structure
e) Condensed structure

(x) ……..is the general term used to explain a mixture of different metals .
a) Alloy
b) Allotrope
c) Amphoteric
d) Amorphous
e) Isotope

2.Match the items in List A with the responses in List B by writing the letter of correct response beside the item number in answer sheet provided

| LIST A | | LIST B |
|---|---|---|
| | | A.Chlorine |
| i) | A gas that forms reddish brown fumes when comes in contact with air | B. Carbonmonoxide<br>C. Nitrogenmonoxide |
| ii) | Vital for all living things | D. Nitrogendioxide |
| iii) | A gas that turn lime water milky | E. Hydrogen |
| iv) | A colourless gas that bleaches moist coloured flowers | F. Nitrogen<br>G. Iodine |
| v) | Only basic gas | H. Hydrogensulphide |
| vi) | A gas that explosive mixture with air and water is only product of combustion | I. Sulphurdioxide<br>J. Bromine |

| | |
|---|---|
| | K. Ammonia<br>L. Carbondioxide<br>M. Flourine<br>N. Oxygen |

**SECTION B: 54 Marks**

3 a) What are the particles that conduct electric current in

i) Sodium metal

ii) Molten sodium chloride

b) Explain why solid sodium chloride would not conduct electric current?

c) An electric current was passed in series through solutions of calcium chloride and copper(ii)sulphate. Carbon electrodes were used in both electrolytes. If 2.5 litres of chlorine measured at STP, what volume of oxygen would also be produced? What mass of copper was also be produced?

4.a) Give five alternative sources of energy.

b) Explain the environmental effects of using charcoal as a fuel (Four points).

5 a) With help of chemical equations where necessary explain the following obsersvations

i) A glass rod which was dipped in a concentrated hydrochloric acid is introduced into a gas jar containing ammonia

ii) A colourless gas which burn with a POP sound it evolved when a piece of magnesium ribbon is droped into a beaker full of dilute hydrochloric acid.

b) Give an account for the following

i) Anhydrous copper(ii)sulphate become coloured when exposed to the air for a long time

ii) Sodium metal is kept under paraffin

iii) A burning splint is introduced to a gas jar containing carbondioxide it goes off

iv) Chlorine gas is collected by downward displacement of air.

6. During the manufacture of sulphuric acid, sulphur trioxide gas is dissolved in concentrated sulphuric acid instead of dissolving it directly in water

(a)Explain why ? write balanced equation of dissolution of the gas in the acid

(b)write a balanced chemical equation for each of the following

i) Sulphuric acid reacting as acid

ii) Sulphuric acid reacting as an oxidizing agent.

7. (a) MR Mwalubhange bought all necessary building materials as he wanted to build modern and expensive house for his mother called nyemba ,Four months later he found out that almost all nails and iron sheets have been rusted as she ignored leakage and small holes in the storage room unfortunately he has no extra money to buy new nails and iron sheet .As a chemistry student what will you advice MR Mwalubhange

(b)suppose you are in a bus travelling from tukuyu for annual holiday ,a bus driver went to the petrol station to fill fuel but he forget to switch off the bus engine ,this caused explosion and fire stated .As a form four student;

i) Will you be able to use fire extinguisher to fight against fire?

ii) Which type of extinguisher will suitable for you to use? Give reason of your choice.

8. a) Compound M contains 37.89% carbon , 4.79% hydrogen and 57.32% oxygen. If the vapour density of M is 30. Calculate the molecular formula of compound M.

b) Two liquids A and B were mixed together but after thoroughly shaking and allowed to settle, liquid B floats.

i) What type of liquid does a mixture A and B form?

ii) Which is the suitable apparatus can be used to separate the two liquids?

**SECTION C: 30 Marks**

9. (a) i) Describe how you can extract sodium metal.

ii) Write down the cathode, anode, and overall reaction taking place in 9(a) (i).

iii) Explain why sodium metal is not used in making bridge.

(b) (i) What is the function of calcium chloride during extraction of that metal.

(ii) Does calcium compete to be liberated at cathode with metal extracted in 9a (i) above? Why?

10. Janet is a form four student who is among the chemistry students at mtakuja secondary school. She failed to attempt the following chemistry questions. You as a form four student help her to answer them.

a) How can you differentiate between acid and base? Give out two points.

b) What is neutralization? Give one example.

c) Is the reaction of acid with carbonate considered as neutralization reaction? Why?

d) The reaction of acids and bases (alkali) are very important in our daily life. With five reasons give the importance of this reaction in our daily life.

11. a) explain the following terms as used in organic chemistry.

i) Isomerism

ii) Homologous series

iii) Functional group

b) Give the IUPAC names of each of the following compounds

i) CH3COOCH3

ii) CH2CH2

iii) CH3CHCHCH3

c) Write the structures of alkanes with the following names

i) 3-methylpentane

ii) 2, 3-dimethylbutane

iii) 2,2-dimethylpropane

# MARKING SCHEME